YouBike Demand Around MRT Stations

R for Data Science Final Project

Group 3

June 7, 2026

Variable Notation

- Y_{ist}: YouBike trip count
- $Rain_{ist}$: Rain indicator
- $Temp_{ist}$: Temperature
- $Humidity_{ist}$: Relative humidity
- $Weekend_s$: Weekend indicator
- $Hour_t$: Hour fixed effects
- $Station_i$: MRT station fixed effects
- ε_{ist}: Error term

$i =$ MRT station $\quad s =$ date $\quad t =$ hour

Model 0: Weather Only

$$Y_{ist} = \beta_0 + \beta_1 Rain_{ist} + \beta_2 Temp_{ist} + \beta_3 Temp^2_{ist} + \beta_4 Humidity_{ist} + \varepsilon_{ist}$$

Model 1: Weather + Time

$$Y_{ist} = \beta_0 + \beta_1 Rain_{ist} + \beta_2 Temp_{ist} + \beta_3 Temp^2_{ist} + \beta_4 Humidity_{ist} + \beta_5 Weekend_s + \sum_h \gamma_h Hour_{ht} + \varepsilon_{ist}$$

Purpose

Examine whether temporal patterns improve explanatory power.

Model 2: Weekend $\times$ Hour

$$Y_{ist} = \beta_0 + \beta_1 Rain_{ist} + \beta_2 Temp_{ist} + \beta_3 Temp^2_{ist} + \beta_4 Humidity_{ist} + \beta_5 Weekend_s + \sum_h \gamma_h Hour_{ht} + \sum_h \delta_h (Weekend_s \times Hour_{ht}) + \varepsilon_{ist}$$

Purpose

Test whether weekday and weekend exhibit different intraday demand patterns.

Model 3: Rain × Hour

$$Y_{ist} = \beta_0 + \beta_1 Rain_{ist} + \sum_h \delta_h (Rain_{ist} \times Hour_{ht}) + \beta_2 Temp_{ist} + \beta_3 Temp_{ist}^2 + \beta_4 Humidity_{ist} + \beta_5 Weekend_s + \varepsilon_{ist}$$

Purpose

Test whether the impact of rain varies across different hours.

Model 4: Station Fixed Effects

$$Y_{ist} = \beta_0 + \beta_1 Rain_{ist} + \beta_2 Temp_{ist} + \beta_3 Temp_{ist}^2 + \beta_4 Humidity_{ist} + \beta_5 Weekend_s + \sum_h \gamma_h Hour_{ht} + \sum_j \theta_j Station_{ij} + \varepsilon_{ist}$$

Purpose

Control for station-specific demand differences.

Model 5: Rain × Weekend

$$Y_{ist} = \beta_0 + \beta_1 Rain_{ist} + \beta_2 Weekend_s + \beta_3 (Rain_{ist} \times Weekend_s) + \beta_4 Temp_{ist} + \beta_5 Temp_{ist}^2 + \beta_6 Humidity_{ist} + \sum_h \gamma_h Hour_{ht} + \sum_j \theta_j Station_{ij} + \varepsilon_{ist}$$

Purpose

Test whether the impact of rain differs between weekdays and weekends.